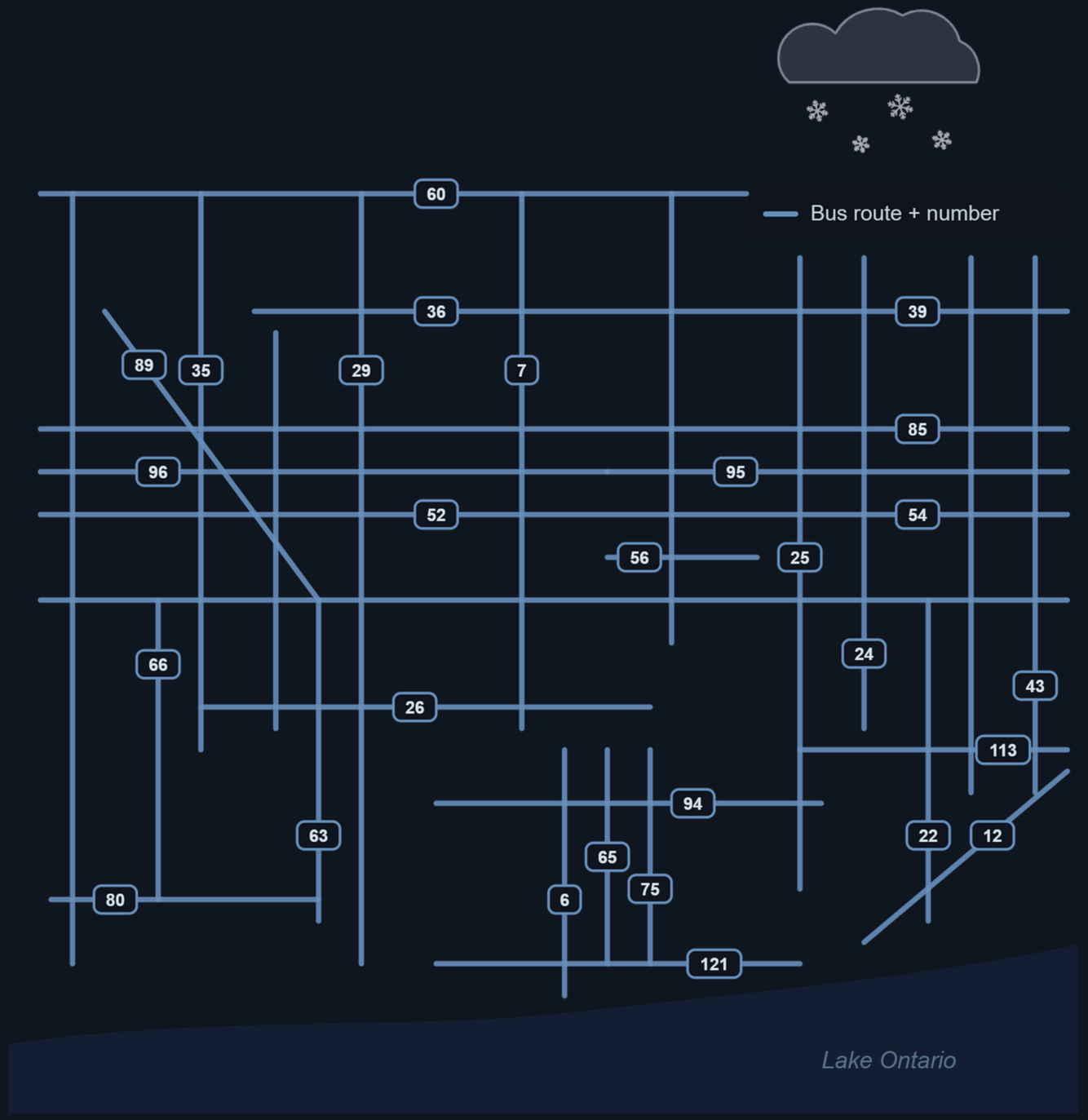
Predicting the effects of extreme precipitation events on TTC operations

How extreme precipitation affects TTC bus delays across Toronto

Team Loon

November 5, 2026



The Engineering Problem

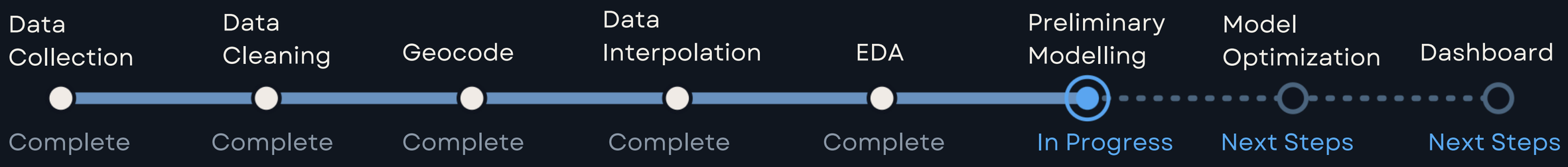
Extreme precipitation events are becoming more common in Toronto, and they disrupt TTC bus service. Riders have no way to know **where** and **how badly** the weather will delay their trip.

Project Objective

1) Predict where and how much TTC bus service will be delayed by precipitation.

2) Show it on a dashboard fed by live weather so riders and TTC operators can plan their trips and routes.

Project Pipeline and Current Project Status



Done

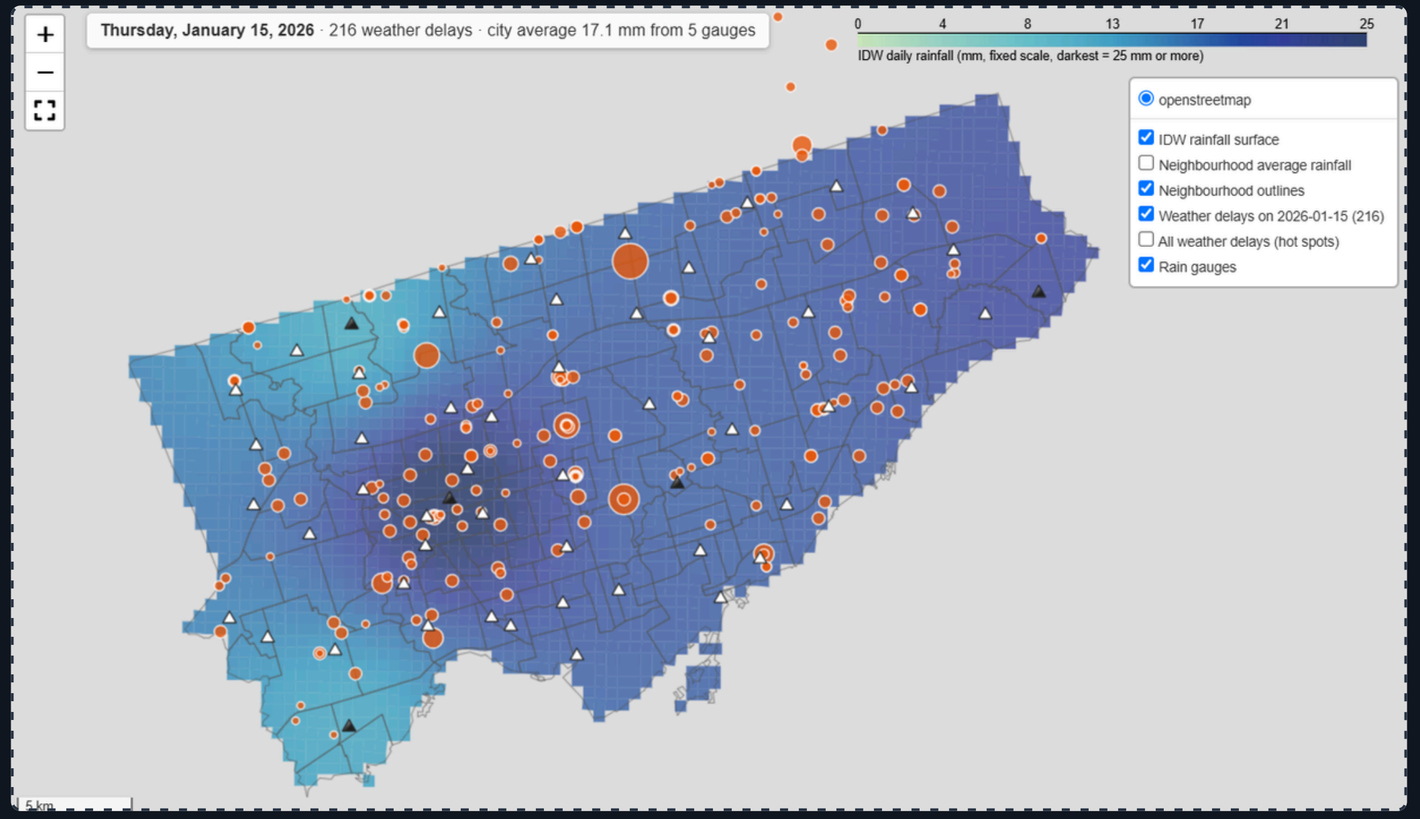
- Rain gauge and bus delay data cleaned and geocoded
- Toronto boundary added to the map

In progress

- Prediction model with 5-fold cross-validation
- Interactive dashboard

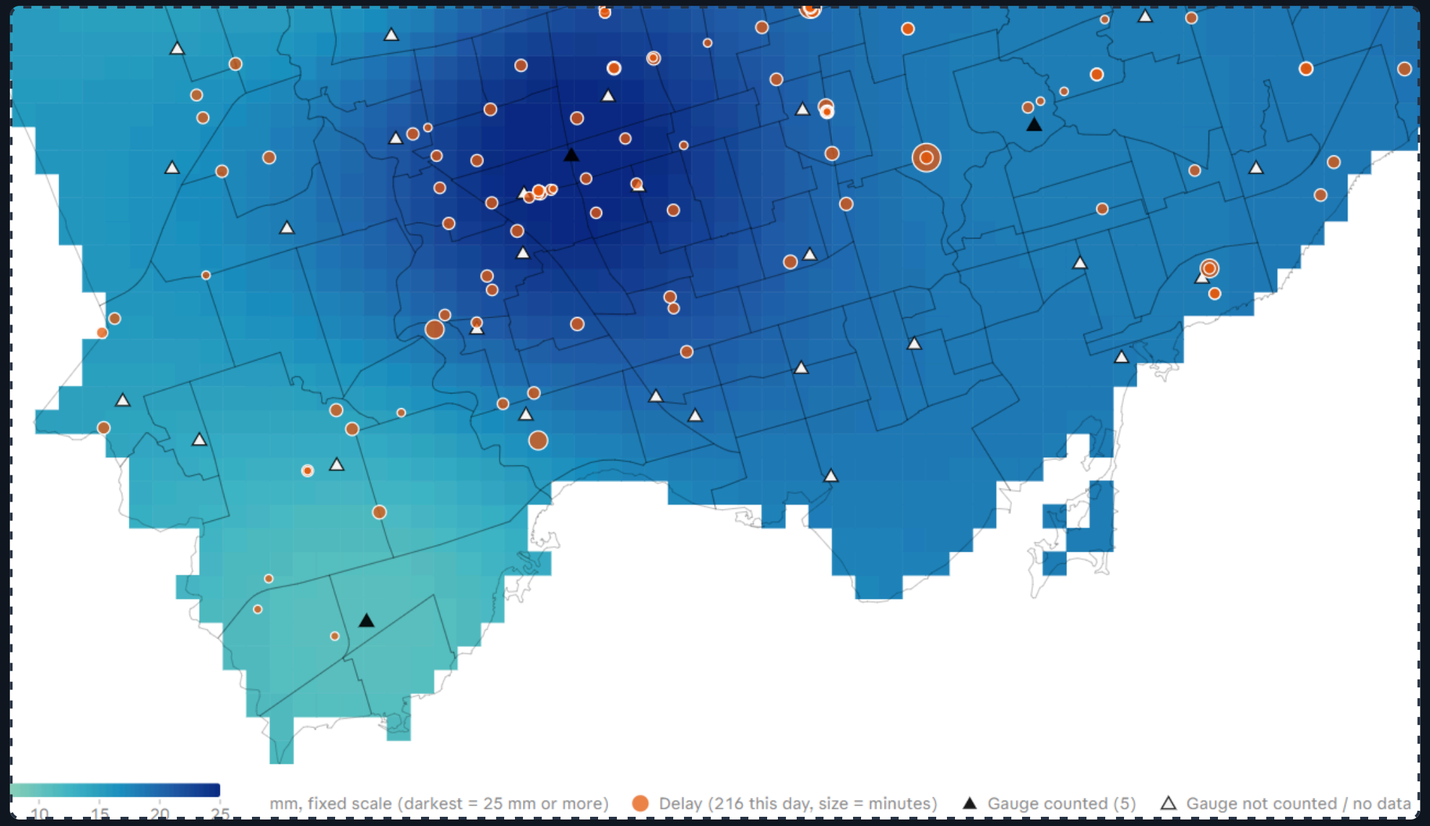
Major progress since D2

- Superimposed TTC delays onto precipitation map.



Major progress since D2

- Created an interactive dashboard to see precipitation and delays on any day between June 2015 to the present 2026.



Major progress since D2

- Training Preliminary Model
  - Linear regression on historical data (from 2015-2026) using 5-fold cross-validation to compare approaches.



Confirming Weather-Related Delays in Pre-2025 Data

2015–2024 delays

No weather label: only General Delay, Operations, Diversion, etc. After 2024 Weather Related Delays are labelled in the dataset.

Service-alert tweets

Scraped TTC Service Alert Tweets for any mention of “weather”, “flood”, “storm” and matched via route and time to delay entries. Removed false postives such as “Four Winds Road”

Add to training set

Over-rode vague labels such as “General Delay” with “Weather Related” to match 2025/6 data

1403
Weather related tweets

861
pre-2025 delays labelled



Data sources used

Source	Provider	Period	Resolution	Key fields
Rain gauges	City of Toronto Open Data (CSV)	2015–2026	5-minute, 44 gauges	Gauge ID, location, precipitation (mm)
TTC bus delays	City of Toronto Open Data (API)	2014–2026	Per delay	Date, time, route, location, incident, min delay
TTC GTFS	City of Toronto Open Data (API)	Current	Stops and routes	Stop coordinates (stops.txt), route shapes
Live weather	Open-Meteo API	Live / forecast	Per coordinate	Precipitation [temperature, wind speed]

Note: Weather is only listed as a delay cause from 2025. Earlier years use General Delay, Operations and Diversion as stand-ins.

Data preparation steps

1. **Standardize.** Unified column names that changed across years and converted all dates to datetime.
2. **Filter.** Kept General Delay, Operations and Diversion as weather proxies; from 2025 we use the TTC's own weather code.
3. **Geocode.** Matched locations to GTFS stops (about 80%), used AI-assisted matching for typed-in intersections, and placed route-wide delays at the route midpoint.
4. **Split gauges by season.** All gauges from April to October; only the 5 heated gauges from November to March.
5. **Interpolate rainfall.** Daily totals estimated at each delay location (Thiessen now, testing IDW), clipped to the Toronto boundary, EPSG:4326.

44
rain gauges: 39 three-season, 5 heated

~80%
of delay locations matched directly to GTFS stops

2133
Confirmed weather-related delay records left after cleaning

Models and analytical methods

Method	Purpose	Training data	Metric	Result
Regression (data subset)	Test how rainfall relates to delay size	Rainfall + delay records	[R^2 / MAE]	25/35/85
Random Forest	Predict whether a delay occurs at a location	Daily rainfall per location, season	[R^2 / MAE]	40/67/86

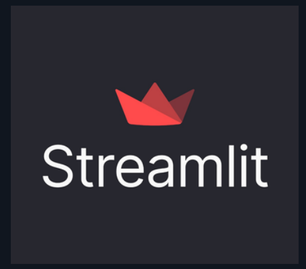
Software and application components

Dashboard (Options Explored):

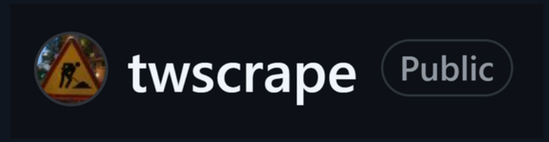
Figma



Streamlit



X Scraping:



Agents:

Claude



Live demo

1 **EDA**
Location and cause of most delays on the TTC

2 **Interactive precipitation and delay map**
Pick any day in 2015–2026, then pan around Toronto

3 **Validation**
Model predictions vs actual delay locations, with accuracy and precision

Remaining work toward final submission

Remaining tasks

- Finish 5-fold model with equal summer/winter shares
- Add temperature and wind speed
- Clean streetcar and subway delays

Planned improvements

- Compare Thiessen vs IDW
- Hourly instead of daily rainfall
- Live forecasts in the dashboard
- Add temperature and wind speed
- Live Open-Meteo weather in the dashboard

Current challenges

- Route-wide and garage-located delays
- Weak labels before 2025
- IDW compute time

Expected outcomes

- Dashboard predicting delays from forecast rain
- Stops that most need precipitation resilience

Questions?

Team Loon: Dana Bou Saab, Connie Zhang, Michelle DeLoyde and Nikita Chiduku

Dashboard: https://torontobusprecipitationdelay.streamlit.app/

Repo: https://github.com/connieyqzhang/CME538-F26-Team05-TeamLoon